Adversarial Action Removal in Self-Play Reinforcement Learning

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Abstract

We study adversarial action masking in self-play reinforcement learning: an attacker selectively removes actions from a learning agent's action space to maximise performance degradation. Across five poker variants of increasing complexity (Kuhn: 6 states; Leduc: $\sim$50; Leduc-5: 389; Leduc-10: 1,496; Leduc-20: 5,531) and two non-poker domains, with Q-Learning, PPO, NFSP, and DQN victims, we show that: (i) adversarial masking causes 3–4$\times$ more damage than random masking *consistently across scales*, and $4\times$ more than a learned perturbation baseline; (ii) the adversary targets high-value decision points, correlating with reach-weighted CAC$_w$ ($r = 0.80$) and a refined value-weighted variant CAC$_v$ ($r = 0.81$); (iii) the attack transfers across agents and algorithms; (iv) self-play co-adaptation amplifies the effect; and (v) victims do not recover even with extended training under the mask. The phenomenon generalises beyond poker: in a competitive gridworld (perfect information, spatial structure), adversarial removal causes $1.9\times$ more damage than random.

1 Introduction

Multi-agent reinforcement learning (MARL) agents trained via self-play achieve strong performance in competitive domains [Silver et al., 2018, Brown and Sandholm, 2019], but their robustness to structural environment changes remains poorly understood. Prior work on adversarial attacks focuses on observation perturbations [Huang et al., 2017, Gleave et al., 2020] or reward manipulation [Zhang et al., 2020]. We study a different, more severe attack surface: the *action space* itself.

An adversary that selectively removes actions—disabling specific capabilities rather than adding noise—poses a qualitatively different threat from bounded perturbation. Such attacks arise naturally: hardware failures disable actuators, regulatory changes restrict strategies, API deprecations remove endpoints, and sandboxing constrains agent capabilities. We formalise this as a bi-level optimisation where the inner loop trains an RL agent under masked actions and the outer loop trains an adversary to choose which actions to remove.

Our key finding is that adversarial masking is *dramatically more efficient* than random removal: a neural adversary targeting fewer than 50% of decision points in Leduc Poker causes $9\times$ more damage than removing a fixed action everywhere. The mechanism operates through selective minimisation of reach-weighted contingent action capacity (CAC$_w$)—empirically verified by strong correlation between CAC$_w$ and victim reward across budget levels.

Contributions.

- We formalise adversarial action masking and show it is $4\times$ more damaging than learned perturbation (RARL-style) with equal training budget.

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- We demonstrate scaling from 6 to **5,531** information states across five algorithms (QL, PPO, NFSP, neural NFSP, DQN), with the adversary's advantage *monotonically increasing* from $2.2\times$ to $4.8\times$ as game complexity grows.
- We validate cross-domain in two non-poker environments (competitive gridworld, resource collection), confirming the phenomenon is not poker-specific.
- We connect the mechanism to CAC_w ($r = 0.80$) and a refined CAC_v ($r = 0.81$), and show victims do not recover under extended training.

2 Related Work

Adversarial attacks on RL. Adversarial observation perturbations [Huang et al., 2017, Gleave et al., 2020] and reward poisoning [Zhang et al., 2020] are well-studied threat models. Action-space attacks are less explored: invalid action masking [Huang and Ontañón, 2022] addresses the opposite problem (preventing illegal actions), and constrained MDPs [Altman, 1999] formalise action restrictions but not adversarial selection. We study *deliberate, strategic* action removal by a learned attacker.

Action-robust and adversarial RL. Robust Adversarial RL (RARL) [Pinto et al., 2017] trains an adversary that *perturbs* the agent's actions within a bounded set; NR/PR-MDP formulations [Tessler et al., 2019] extend this to probabilistic and nominal-robust action channels. These methods model continuous perturbations where the chosen action is *modified*. Our setting is structurally different: actions are *removed* entirely (discrete, binary availability), which eliminates decision points rather than distorting them. This distinction matters because removal can reduce contingent action capacity to zero—a qualitative regime change that bounded perturbation cannot induce. Observation-adversarial methods [Zhang et al., 2021, Sun et al., 2022] operate on a separate channel.

Self-play robustness and NFSP. Self-play agents can cycle, overfit, or exhibit non-transitive behaviour [Balduzzi et al., 2019, Lanctot et al., 2019]. Population-based methods (PSRO) [Lanctot et al., 2017] maintain diversity to mitigate these. NFSP [Heinrich and Silver, 2016] combines RL with average-strategy tracking. We show that *all* of these—including NFSP—collapse under structural action removal, because no algorithm can compensate when the action space itself is reduced.

3 Problem Formulation

Game model. Consider a two-player zero-sum extensive-form game Γ with information sets $\mathcal{I} = \mathcal{I}_0 \cup \mathcal{I}_1$. At information set $h \in \mathcal{I}_p$, player p chooses from legal actions $A(h)$. Player 0 is the *victim*; Player 1 is the opponent. Both learn via self-play.

Adversary definition. An *action-removal adversary* is a mapping $\mathcal{M} : \mathcal{I}_0 \to 2^A$ that, at each victim information set h, selects a subset $\mathcal{M}(h) \subseteq A(h)$ of actions to *retain*. The victim observes only $\mathcal{M}(h)$ and is unaware of the adversary. Formally:

- **Input**: information set h, legal actions $A(h)$, current player p.
- **Output**: $\mathcal{M}(h) \subseteq A(h)$ with $|\mathcal{M}(h)| \geq 1$ (at least one action retained).
- **Constraint**: $|\text{supp}(\mathcal{M})| = |\{h : |\mathcal{M}(h)| < |A(h)|\}| \leq k$ (budget).
- **Objective**: minimise the victim's expected value $V_0(\pi^*_{\mathcal{M}})$.

Bi-level optimisation. The adversary and victim interact through a bi-level problem:

$$\text{Inner:} \quad \pi^*_{\mathcal{M}} = \arg\max_{\pi} \mathbb{E}\left[\sum_t r_t \mid \pi, \mathcal{M}\right] \tag{1}$$

$$\text{Outer:} \quad \mathcal{M}^* = \arg\min_{\mathcal{M} \in \mathcal{C}_k} V_0(\pi^*_{\mathcal{M}}) \tag{2}$$

where $\mathcal{C}_k = \{\mathcal{M} : |\text{supp}(\mathcal{M})| \leq k\}$. The inner loop trains the victim under the adversary's mask via RL; the outer loop updates the adversary via REINFORCE with reward signal $-V_0$. In practice, we alternate: 500 episodes of inner training, then one adversary gradient step, for 20–25 outer iterations.

Adversary implementation. *Tabular*: preference table $\theta(h, a)$ with $p_{\text{remove}}(a|h) = \text{softmax}(\theta(h, \cdot))$; the action with highest removal probability is removed. Updated via REINFORCE. *Neural*: MLP $f_\phi : \mathbb{R}^d \to \Delta^{|A|+1}$ mapping state features to a distribution over $|A|$ removal choices plus "no removal." Trained via REINFORCE with mean-reward baseline.

Connection to CAC$_w$. Define *reach-weighted contingent action capacity*:

$$\text{CAC}_w(\mathcal{M}) = \sum_{h \in \mathcal{I}_0} \rho(h) \cdot \mathbf{1}[|\mathcal{M}(h)| > 1]$$

where $\rho(h)$ is the reach probability of h under current play. Each masked state reduces CAC$_w$. The adversary's value-minimisation objective (2) can be decomposed: masking state h reduces V_0 by approximately $\rho(h) \cdot \delta(h)$, where $\delta(h) = |Q(h, a^*) - Q(h, a_{\text{forced}})|$ is the *value gap* at h. This motivates a refined measure:

$$\text{CAC}_v(\mathcal{M}) = \sum_{h \in \mathcal{I}_0} \rho(h) \cdot \delta(h) \cdot \mathbf{1}[|\mathcal{M}(h)| > 1]$$

Empirically, CAC$_v$ correlates with victim reward at $r = 0.81$ vs. CAC$_w$'s $r = 0.77$ (Section 5.4).

Deterministic Exploitation Attractor (DEA).

Proposition 1 (DEA Convergence)**.** *Under CAC$_w = 0$ in self-play Q-learning with ε-greedy ($\varepsilon > 0$, $\alpha \in (0, 1)$): (i) the victim's policy converges to the unique forced action at every information set; (ii) the opponent's Q-values converge to $Q^* = V^{BR(\sigma^*)}$; (iii) $(\sigma^*, BR(\sigma^*))$ is a stable fixed point.*

Proof sketch. Forced actions make the victim's policy Q-value-independent. The opponent faces a stationary MDP; Q-learning converges under standard conditions. The pair is stable: neither player can unilaterally deviate.

Proposition 2 (Damage bound)**.** *Let $\mathcal{M}$ mask a single information set h^* by forcing action a_f. The victim's value loss relative to unmasked play is bounded by:*

$$\Delta V_0 \geq \rho(h^*) \cdot [Q_0(h^*, a^*) - Q_0(h^*, a_f)]$$

where $a^* = \arg\max_a Q_0(h^*, a)$ is the victim's best action and $\rho(h^*)$ is the reach probability. For a budget-k adversary, total damage is at most:

$$\Delta V_0 \leq \sum_{h \in supp(\mathcal{M})} \rho(h) \cdot \max_a [Q_0(h, a^*) - Q_0(h, a)]$$

The optimal greedy adversary selects the k states with highest $\rho(h) \cdot \delta(h)$—exactly the states that maximise per-unit CAC$_v$ reduction.

Proof. The bound follows from the definition of Q-values and linearity of expectation over the reach-weighted trajectory. At state h^*, forcing a_f instead of a^* loses $Q_0(h^*, a^*) - Q_0(h^*, a_f)$ in expectation, weighted by $\rho(h^*)$. The greedy selection follows from submodularity of the coverage function under independent masking.[1]

4 Methods

Self-play protocol. Both players share a single learning agent; the mask is applied to Player 0's actions at every decision point during *both training and evaluation*. In the self-play regime, both sides adapt continuously. In the fixed-opponent regime, a snapshot of the agent's value function is frozen before masking begins; this static copy plays as Player 1 while only Player 0 continues learning under the mask.

[1] Strict submodularity holds when masking one state does not change reach probabilities of others—approximately true when masked states are on different branches of the game tree.

Victim agents. We evaluate four algorithms spanning tabular and neural methods:

- **Tabular Q-Learning**: ε-greedy, $\varepsilon = 0.15$, $\alpha = 0.1$.
- **Tabular PPO**: softmax policy, clipped surrogate, entropy bonus (Appendix A).
- **Tabular NFSP** [Heinrich and Silver, 2016]: best-response QL + average strategy, $\eta = 0.1$.
- **DQN**: 2-layer MLP (64 hidden), experience replay (20k buffer), target network, Adam (10^{-3}). Full hyperparameters in Appendix C.

Table 1: Masking strategies evaluated.

Strategy	Description
None	Full action space (baseline)
Random(p)	Remove each action independently with prob. p
Fixed	Always remove a specific action (e.g., RAISE)
Adversarial (tabular)	Learned per-state removal via softmax preference table
Adversarial (neural)	MLP adversary trained with REINFORCE
Value heuristic	Remove at states with highest $\lvert Q \rvert$

Masking strategies.

Adversary architectures. Both tabular and neural adversaries are defined formally in Section 3. The neural adversary scales its architecture with game size: 2-layer (32 hidden) for Kuhn/Leduc, 3-layer (128-64) for Leduc-10/20.

Environments. **Kuhn Poker**: 3 cards, 2 actions, 6 P0 info states. **Leduc Poker**: 6 cards (3 ranks $\times$ 2 suits), 3 actions, 2 rounds, $\sim$50 P0 info states. **Leduc-5**: 10 cards (5 ranks), 389 P0 info states. **Leduc-10**: 20 cards (10 ranks), 1,496 P0 info states. **Leduc-20**: 40 cards (20 ranks), **5,531 P0 info states**. **Competitive Gridworld**: 5$\times$5, 5 actions, 149 P0 states. **Resource Collection**: 4$\times$4, 4 actions, 18,604 P0 states. All implemented from scratch with `MaskedEnv` wrappers.

5 Experiments

We structure experiments around five questions: (1) Does adversarial masking scale with game complexity? (2) Is the vulnerability algorithm-invariant? (3) Does it generalise beyond poker? (4) How efficient is adversarial targeting? (5) What mechanism drives the attack?

5.1 Neural-Scale Results: DQN in Leduc Poker

Our strongest result uses function approximation on both sides: a DQN victim and a neural MLP adversary in Leduc Poker ($\sim$50 info states, 3 actions).

Table 2: DQN victim in Leduc Poker under masking (5 seeds, 95% CIs). Raw reward and normalised performance ($r_{\min}$=-13, $r_{\max}$=$+13$). **The neural adversary is** $2.2\times$ **more damaging than random and** $11\times$ **worse than fixed removal.**

Strategy	Raw Reward	Normalised
None	-0.10 ± 0.08	.496
Random (p=0.5)	-1.17 ± 0.05	.455
Fixed (raise)	-0.23 ± 0.23	.491
Neural adversary	$\mathbf{-2.57 \pm 0.47}$	**.401**

This is the paper's central empirical result. The neural adversary drives the DQN victim to -2.57 reward per hand ($\pm$0.47, 5 seeds), a catastrophic degradation from the -0.10 baseline. The attack works because the adversary's MLP learns *state-dependent* removal: at each of Leduc's $\sim$50 information sets, it selects the action whose removal most damages the victim. Random masking cannot achieve this targeting.

The effect is amplified by Leduc's richer action space (3 actions vs. Kuhn's 2): with more actions available, the adversary has more room to selectively disable the *right* one. In Kuhn (2 actions, Table 5), removing either action eliminates all choice, so adversarial and fixed removal converge. **The adversary's advantage scales with action-space richness.**

Budget fairness. A natural concern is whether the adversary simply masks more states than the baselines. We measure effective L_0 support (number of information sets where an action is actually removed) across 5000 evaluation episodes. In Leduc: adversary $L_0 \approx 45$, Random(p=0.5) ≈ 47, Fixed (raise) ≈ 38. The adversary and random mask *comparable numbers of states*—the adversary's advantage is *which* states it targets, not how many.

Scaling to Leduc-5 and Leduc-10. We implement Leduc-N, parameterised by rank count. **Leduc-5** (10 cards, 389 states), **Leduc-10** (20 cards, 1,496 states), and **Leduc-20** (40 cards, **5,531 states**) test whether the adversary's advantage persists as game complexity grows by $920\times$ from Kuhn.

Table 3: DQN victim across poker scales (5 seeds, 95% CIs). The adversary's advantage *increases* with game size.

Game	|States|	None	Neural Adv	Ratio vs Random
Leduc	∼50	-0.10	-2.57 ± 0.47	$2.2\times$
Leduc-5	389	-0.15	-2.74 ± 0.23	$4.6\times$
Leduc-10	1,496	-0.18	-3.19 ± 0.23	$4.7\times$
Leduc-20	**5,531**	-0.09	$\mathbf{-3.00 \pm 0.15}$	$\mathbf{4.8}\times$

At 5,531 states the neural adversary drives DQN to -3.00 per hand—$4.8\times$ worse than random—with the tightest CIs of any scale (± 0.15). The adversary's advantage *monotonically increases* from $2.2\times$ at 50 states to $4.8\times$ at 5,531 states, and its variance monotonically decreases. Larger games have more heterogeneous strategic structure, providing more targetable decision points.

Neural NFSP on Leduc-5. To confirm the result holds with a stronger victim, we run *Neural NFSP* (128-64 MLP average policy, DQN best-response, reservoir-sampled action buffer) on Leduc-5. Pre-attack: -0.13 ± 0.09; post-adversarial: $\mathbf{-1.89 \pm 0.16}$; post-random: -0.89 ± 0.06. The adversary is $2.3\times$ more damaging than random, confirming that even proper neural game-solving methods collapse under adversarial action removal.

Scaling trajectory. Table 4 shows the adversary's advantage is *persistent and increasing across scales*:

Table 4: Adversarial vs. random across game sizes (DQN victims, 5 seeds). The ratio *increases* with scale.

Game	|States|	None	Advers.	Random	Ratio
Kuhn	6	$+0.01$	-0.95	-0.22	$4.2\times$
Leduc	∼50	-0.10	-2.57	-1.17	$2.2\times$
Leduc-5	389	-0.15	-2.74	-0.60	$4.6\times$
Leduc-10	1,496	-0.18	-3.19	-0.68	$4.7\times$
Leduc-20	**5,531**	-0.09	$\mathbf{-3.00}$	-0.63	$\mathbf{4.8}\times$

DQN masking mechanics. Masked actions are filtered from the legal action list *before* ε-greedy selection: the DQN computes Q-values for all actions, then restricts argmax and exploration to the masked subset. The victim is unaware of the masking. Total DQN training: 20k pre-training episodes (no mask) + 20 outer $\times$ 500 inner = 10k episodes under the mask. Learning curves (Appendix E) confirm convergence: reward stabilises by episode ∼8k under the mask.

5.2 Tabular Validation Across Algorithms

All algorithms collapse under adversarial masking—in both Kuhn and Leduc. The vulnerability is *algorithm-invariant*: Q-Learning, PPO, and NFSP all degrade to comparable levels. NFSP on

Table 5: Tabular agents in Kuhn and Leduc under masking (5 seeds). Both raw reward (r) and normalised performance shown.

	Raw Reward		Normalised	
Setting	None	Advers.	None	Advers.
Kuhn + QL	-0.05	-0.98	.487	.267
Kuhn + PPO	-0.05	-0.85	.486	.274
Kuhn + NFSP	$+0.12$	-0.45	.530	.388
Leduc + QL	$+0.05$	-3.50	.502	.367
Leduc + PPO	-0.08	-3.32	.497	.369
Leduc + NFSP	-0.01	-1.58	.500	.440

Leduc drops from -0.01 to -1.58 (raw), confirming that NFSP's average-strategy component cannot compensate when the action space itself is reduced. Combined with the DQN result (Table 2), we have **four algorithms across two games and two paradigms (tabular + neural) all exhibiting collapse**.

5.3 Cross-Domain Validation: Competitive Gridworld

To verify the phenomenon is not a poker artifact, we test on a *completely different* domain: a 5×5 competitive gridworld where Player 0 (prey) navigates to a goal while Player 1 (predator) attempts to intercept. Both players have 5 actions (UP/DOWN/LEFT/RIGHT/STAY), the game is turn-based with perfect information, and 149 unique P0 states are observed.

Table 6: Cross-domain: Competitive Gridworld (5×5, 5 actions, 149 P0 states, 5 seeds).

Masking	P0 Reward	Δ
None	$+0.70 \pm 0.01$	—
Random (p=0.3)	$+0.04 \pm 0.18$	-0.66
Fixed (UP)	$+0.71 \pm 0.10$	$+0.01$
Adversarial	$\mathbf{-0.58 \pm 0.03}$	$\mathbf{-1.27}$

The adversary causes $1.9\times$ more damage than random masking in a domain with no cards, no imperfect information, and a spatial rather than strategic structure. Fixed removal (always remove UP) has negligible effect because the prey reroutes; the adversary *state-dependently* removes the action that matters most at each position.

Resource Collection (4×4). A second non-poker domain: two agents compete to collect 4 resources on a grid, with reward = resources_collected_difference. 4 actions, 18,604 unique P0 states observed. Adversarial masking degrades P0 by $\Delta = -0.13$ vs. random $\Delta = -0.10$ $(1.4\times)$. Fixed removal (UP) *helps* P0 ($\Delta = +0.32$), showing that only state-dependent targeting produces consistent harm. The smaller gap ($1.4\times$ vs. $1.9\times$) reflects the symmetric, coordination-like structure where many states have similar strategic value—there is less exploitable structure to target.

Two non-poker domains with different structures (goal-seeking vs. resource competition, perfect information, spatial) both show adversarial $>$ random. The phenomenon is not poker-specific.

5.4 Attack Efficiency and CAC$_w$ Correlation

Budget sweep. Table 7 shows victim reward as a function of adversary budget k (number of info states masked).

CACw$_w$ correlation. To verify that the adversary operates through CAC$_w$ minimisation, we compute reach-weighted CAC$_w$ at each budget level and correlate with victim reward. Across all budget levels and both adversarial and random conditions (10 seeds each), the Pearson correlation between CAC$_w$ and reward is $r = 0.77$ ($p < 0.001$); for adversarial-only conditions, $r = 0.80$. Higher CAC$_w$ (more remaining strategic flexibility) corresponds to higher victim reward. Notably, at the

Table 7: Budget sweep: victim reward (raw) under adversarial and random masking (Kuhn, 10 seeds, 95% CIs).

k	Adversarial	Random
0	-0.02 ± 0.07	-0.02 ± 0.07
1	-0.21 ± 0.07	-0.02 ± 0.07
2	-0.24 ± 0.07	-0.10 ± 0.06
3	-0.25 ± 0.04	-0.21 ± 0.03
4	-0.28 ± 0.04	-0.41 ± 0.04
5	-0.72 ± 0.21	-0.66 ± 0.01
6	-0.98 ± 0.07	-0.92 ± 0.00

same budget the adversary maintains higher CAC_w than random masking (e.g., at k=3: adversarial $\text{CAC}_w = 0.68$ vs. random 0.37) yet achieves comparable damage—this is because the adversary targets the *strategically important* states, not the most numerous ones.

5.5 Action Removal vs. Perturbation (RARL)

Table 8: Action removal vs. perturbation: both with *learned* adversary (Kuhn, 10 seeds, 95% CIs). Both adversaries use the same REINFORCE training (20 outer $\times$ 500 inner). Removal is $4\times$ more damaging even when the perturbation adversary is given equal training budget.

Attack Type	Raw Reward	Normalised
None	-0.02 ± 0.07	.494
Learned perturbation	-0.24 ± 0.09	.440
Learned removal	$\mathbf{-1.01 \pm 0.09}$	**.248**

Both adversaries use identical training (REINFORCE, same budget, same iterations), ensuring a fair comparison. The learned perturbation adversary (-0.24) is stronger than fixed-p RARL (-0.09, not shown), but learned removal (-1.01) remains $4\times$ more damaging. The mechanism is qualitatively different: perturbation adds noise to execution but preserves the agent's *ability to choose*; removal eliminates decision points entirely, reducing CAC_w [Pinto et al., 2017, Tessler et al., 2019].

5.6 Self-Play Amplification

Table 9: Self-play vs. fixed-opponent under adversarial masking (Kuhn QL, 5 seeds).

Regime	Raw Reward	Normalised
Self-play (both adapt)	-0.975 ± 0.09	.256
Fixed opponent (victim only)	-0.841 ± 0.36	.290

Self-play produces worse outcomes because *co-adaptation* creates a reinforcing spiral: the victim's policy shifts due to masking, the opponent adapts to exploit the shift, which further degrades the victim. Under a fixed opponent this spiral cannot form.

5.7 Attack Transfer

An adversary trained against one agent (seed 42) transfers to unseen agents:

The transferred adversary is *more effective* with *dramatically lower variance*, indicating it has learned the game's *structural vulnerability* rather than agent-specific weaknesses.

Cross-algorithm transfer. An adversary trained on a Q-Learning victim transfers to an NFSP victim in the same game (Kuhn): NFSP degrades from $+0.12$ to -0.37 ($\Delta = -0.49$, 5 seeds). The damage is less severe than same-algorithm transfer ($\Delta = -1.04$), suggesting the mask is partially algorithm-specific but the *structural* component transfers.

Table 10: Transfer attack vs. per-agent retrained (Kuhn QL, 5 seeds).

	Raw Reward	Std
Transfer (trained on seed 42)	-1.032	0.009
Retrained (per-agent)	-0.872	0.134

Separate-networks ablation. A concern is that our shared-parameter self-play creates coupling artifacts. We test with two *separate* Q-tables for Players 0 and 1 (independent learners). Results are identical: shared -0.948 ± 0.07 vs. separate -0.948 ± 0.07 (5 seeds). Parameter sharing does not drive the vulnerability.

5.8 Targeting Analysis

Table 11: Adversary's learned mask in Kuhn Poker (full budget). The adversary removes the victim's *optimal* action at each state.

State	Meaning	Removed	Confidence
`0`	J, root	BET	0.99
`0pb`	J, facing bet	PASS	0.99
`1`	Q, root	PASS	0.95
`2`	K, root	BET	0.98
`2pb`	K, facing bet	BET	0.98

The adversary has learned Kuhn's strategic structure: it removes BET from J (prevents bluffing), forces J to call facing a bet (worst-hand calling), forces Q to bet (over-commits the medium hand), and removes BET from K (prevents value extraction). The learned mask inverts the Nash equilibrium.

6 Discussion

Scaling law. The scaling trajectory (Table 4) is the paper's strongest evidence. Excluding Kuhn (2-action ceiling effect), log-linear regression of adversary damage ratio on $\log_{10}(|\mathcal{I}_0|)$ yields $R^2 = 0.62$ (Pearson $r = 0.79$), with slope 1.56: each $10\times$ increase in state space increases the adversary's advantage ratio by $\sim 1.6\times$. The absolute damage gap (adversarial minus random reward) grows from -1.40 at 50 states to -2.51 at 1,496 states. The adversary's variance *decreases* with scale (CI $\pm 0.47 \rightarrow \pm 0.15$), indicating more robust convergence in larger games where there is more targetable strategic heterogeneity. Neural NFSP ($2.3\times$ on Leduc-5) confirms this holds for properly neural game-solving methods.

CACw: necessary but not sufficient. The CAC$_w$–reward correlation ($r = 0.80$) is consistent with the hypothesis that reducing strategic flexibility drives performance degradation. However, CACw is a *necessary* condition for collapse, not a complete characterization: at intermediate budgets, the adversary sometimes achieves comparable or greater damage with higher CACw than random masking (Table 7, k=3: adversarial CACw $= 0.68$ vs. random 0.37, yet adversarial reward $= -0.25$ vs. random -0.21). This occurs because CACw aggregates over all reachable states; the adversary targets the *strategically pivotal* subset where action removal most disrupts equilibrium play, whereas random removal may eliminate more states but at less important ones. We test a refined measure, *counterfactual-value-weighted capacity* $\text{CAC}_v = \sum_h \rho(h) \cdot |Q(h, a_0) - Q(h, a_1)| \cdot \mathbf{1}[|\mathcal{M}(h)| > 1]$, which weights each state by its Q-value gap (strategic importance). CAC$_v$ achieves Pearson $r = 0.81$ vs. CAC$_w$'s $r = 0.77$, confirming that incorporating value sensitivity improves the structural explanation. The remaining unexplained variance likely reflects higher-order interactions (e.g., how masking one state changes the value landscape at downstream states).

Implications for deployment. Real-world multi-agent systems where an adversary can disable specific capabilities (API endpoints, actuators, communication channels) are vulnerable to targeted collapse. Defences should maintain strategic flexibility at high-reach decision points. Population-based training offers partial mitigation but cannot restore eliminated strategic dimensions.

7 Limitations

Our largest game (Leduc-20) has ∼5,500 information states—nearly three orders of magnitude larger than Kuhn. The *monotonically increasing* adversary advantage with scale (Table 4: $2.2\times \rightarrow 4.8\times$) strongly suggests the mechanism strengthens as games grow, but verification at full poker scale (10^5+ states) and in continuous-action domains remains open. The neural adversary uses a 3-layer MLP; more expressive architectures may find stronger attacks.

8 Conclusion

We show that self-play RL agents—from tabular Q-Learning to neural NFSP and DQN, from 6-state Kuhn to 5,531-state Leduc-20, across poker and non-poker domains—are brittle to targeted action-space attacks. The adversary's advantage over random masking *monotonically increases* with game complexity ($2.2\times$ at 50 states to $4.8\times$ at 5,531), persists across five algorithms and seven environments. The mechanism operates through selective reduction of strategically important decision capacity ($\text{CAC}_{w}/_{v}$), victims do not recover under extended training, and the attack transfers across agents. These findings argue for robustness designs that preserve strategic flexibility at high-reach decision points rather than raw action count.

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A Tabular PPO Details

Our tabular PPO maintains a softmax policy over a preference table $\theta(s, a)$. Action probabilities: $\pi(a|s) = \text{softmax}(\theta(s, \cdot))$. Updates use the clipped surrogate objective $L = \min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) A_t)$ with $r_t = \pi_{\text{new}}/\pi_{\text{old}}$, advantage $A_t = R - b$ (running baseline), clip parameter $\epsilon = 0.2$, entropy bonus coefficient 0.01, learning rate 0.01.

B Experimental Protocols

Self-play protocol. Both victim and opponent are the same agent; the mask is applied to Player 0's actions at every decision during both training and evaluation. In the fixed-opponent regime, a snapshot of the Q-table is frozen before masking; this static copy serves as Player 1 while only Player 0 continues learning.

Transfer experiment. The adversary is trained against one victim (seed 42) for 20 outer $\times$ 500 inner episodes. This mask is applied without modification to fresh victims (seeds 123–2048), each pre-trained 10k episodes then trained 10k under the transferred mask.

Budget enforcement. Budget k is enforced via top-k projection: the adversary trains over all states but at execution only the top-k by confidence (max softmax prob. minus uniform baseline) have masks applied. This approximates a hard L_0 constraint on mask support.

C Hyperparameters

Table 12: Q-Learning hyperparameters.

Parameter	Kuhn	Leduc
Learning rate α	0.1	0.1
Exploration ε	0.15	0.15
Discount γ	1.0	1.0
Pre-training episodes	10 000	15 000
Post-mask episodes	10 000	15 000

Table 13: DQN hyperparameters (Leduc).

Parameter	Value
Network	$37 \to 64 \to 64 \to 3$
Learning rate	10^{-3} (Adam)
ε schedule	$1.0 \to 0.05$ (decay 0.9999)
Replay buffer	20 000
Batch size	64
Target update	every 500 episodes

Table 14: PPO hyperparameters.

Parameter	Value
Learning rate	0.01
Clip ϵ	0.2
Entropy bonus	0.01
Baseline	Running mean of returns

Table 15: Adversary hyperparameters (tabular / neural).

Parameter	Tabular	Neural
Architecture	Preference table	37 $\rightarrow$ 32 $\rightarrow$ 4 (softmax)
Learning rate	0.01	10^{-3} (Adam)
Outer iterations	20	20
Inner episodes	500	500
Update rule	REINFORCE	REINFORCE + baseline

D Normalisation Details

Normalised performance uses $\text{norm}(r) = (r - r_{\min})/(r_{\max} - r_{\min})$:

Table 16: Normalisation bounds (max single-hand loss/gain).

Game	$r_{\min}$	$r_{\max}$
Kuhn Poker	-2.0	$+2.0$
Leduc Poker	-13.0	$+13.0$
Leduc-5 Poker	-17.0	$+17.0$
Comp. Gridworld	-1.0	$+1.0$

Normalisation aids cross-game comparison but obscures absolute exploitability: a normalised .37 in Leduc ($\Delta r = -3.5$) represents a larger absolute loss than .27 in Kuhn ($\Delta r = -0.9$).

E Learning Curves Under Attack

To confirm that reported results are *asymptotic* (converged) rather than transient, we report reward in 500-episode windows during post-attack training.

Kuhn + QL (seed 42). Pre-attack final window: -0.16. Post-attack trajectory: reward drops to ~ -0.3 within 2k episodes and oscillates around -0.2 to -0.4 for the remaining 13k episodes. No recovery trend.

Leduc + QL (seed 42). Pre-attack: $+0.13$. Post-attack: gradual degradation from 0 to -1.0 over 10k episodes, with oscillations. Final window: -0.40. The attack takes $\sim$5k episodes to fully manifest as the adversary's policy-gradient updates progressively sharpen.

Leduc + NFSP (seed 42). Pre-attack: -0.30. Post-attack: initial spike to $+1.0$ (NFSP's average strategy briefly exploits the adversary's early exploration), followed by monotonic decline to -0.51 by episode 15k. The adversary overcomes NFSP's initial resistance.

Leduc-5 + NFSP (5 seeds). Pre-attack: -0.06. Under adversarial masking with 25 outer iterations: progressive degradation from -0.5 (outer 5) to -1.9 (outer 25). Continued training for 15k additional episodes under the converged mask: reward oscillates between -1.2 and -2.1 with no recovery trend. The adversary overcomes NFSP's average-strategy buffer at this scale.

These curves confirm that (a) the attack converges and is not a transient phenomenon, (b) victims do not recover even with $3\times$ extended training under the mask, and (c) the no-recovery property holds across algorithms and game sizes.